

Project 3

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1 Introduction

When I was taking chemistry in high school, I learned about structural isomers. These were molecules that had the same chemical formula, but different structures. The example given was that of butane (C_4H_{10}), which can either be a straight chain of 4 carbon atoms, or a branched chain with 3 carbon atoms in a row and one carbon atom attached to the middle carbon atom (see Figure 1). Butane belongs to a class of molecules called alkanes, which are composed

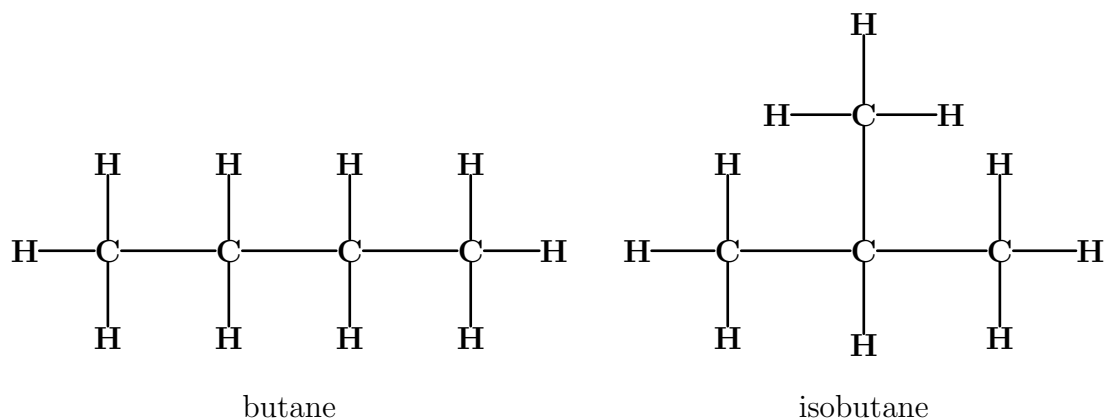


Figure 1: The two structural isomers of C_4H_{10} .

entirely of carbon and hydrogen atoms with only single bonds. Alkanes can have many structural isomers depending on the number of carbon atoms present in the molecule. The number grows very rapidly as you can see from the table in Figure 2. This made me wonder: is there a formula or algorithm that can determine the number of structural isomers for any given alkane? After doing some research, I found out about the Online Encyclopedia of Integer Sequences (OEIS) which has a sequence (A000602) that lists the number of structural isomers for alkanes. It presents a generating function by Eric Rains and Neil Sloane based on Arthur Cayley's work in 1875 [1].

Number of Carbons	Number of Structural Isomers
1	1
2	1
3	1
4	2
5	3
6	5
7	9
8	18
9	35
10	75

Figure 2: Number of structural isomers of alkanes with different number of carbon atoms.

2 The Generating Function

First we will give a few definitions.

- A **graph** is a collection of vertices (or nodes) connected by edges.
- The **degree** of a node is the number of edges connected to it.
- The **valence** of a graph is the maximum degree of any node in the graph. In our case, since carbon atoms can form up to 4 bonds, we will be considering graphs that are 4-valent.
- A **tree** is a connected graph with no cycles. This means there is exactly one path between any two nodes in the tree.
- A **rooted tree** is a tree with one vertex designated as the root, giving the tree a sense of direction away from the root.
- The **diameter** of a tree is the length of the longest path between any two nodes in the tree.
- The **center** of a tree is the node (or pair of nodes) that is in the middle of the longest path in the tree.

3 Bibliography

References

- [1] Eric M Rains and Neil JA Sloane. On cayley's enumeration of alkanes (or 4-valent trees). *Journal of Integer Sequences*, 2(99.1):1, 1999.